

Vaultix Password Manager

Complete Platform & Feature Documentation

1. Platform Overview

Vaultix is an enterprise-grade password management platform designed to deliver end-to-end security for personal and organizational credentials. It enables individuals and teams to securely store, organize, generate, analyze, and share sensitive credentials without sacrificing usability or performance.

Vaultix is built on a Zero-Knowledge architectural model, ensuring that private vault items remain accessible exclusively to authorized account holders. Through a combination of client-side encryption, granular role-based permissions, automated security scoring, and audit logging, Vaultix provides a single glass pane for organization-wide password governance.

2. Key Capabilities & Features

- **Personal & Shared Vaults:** Dual-vault workspace separating private credentials from organizational team credentials.
- **Granular Folder Management:** Multi-level folder hierarchies with flexible permission levels.
- **Individual Credential Sharing:** Direct credential-level sharing with team members and instant access revocation.
- **Zero-Knowledge Security Model:** Dynamic, client-side encryption using derived cryptographic keys.
- **Password Generator Engine:** Comprehensive password generator with custom character set rules and entropy analysis.
- **Password Security Auditor:** Real-time checking against known compromised passwords, weak structures, and old entries.
- **Security Health Dashboard:** Executive dashboard featuring automated security health scores and activity trend graphs.
- **Activity & Audit Logging:** Complete operational audit trails for credential views, copies, edits, shares, and deletions.
- **Login Activity Monitor:** Real-time IP address and browser tracking for authentication attempts with suspicious login alerts.
- **Bulk Excel Import & Export:** Seamless password migration via encrypted Excel spreadsheet import and structured export.
- **Team & User Administration:** Centralized user management, role assignments, user status toggling, and email invitations.

- **Notification System:** Built-in alert center for security warnings, share requests, and account activity.

3. Security & Zero-Knowledge Architecture

Vaultix ensures that user passwords and confidential notes are never stored or transmitted in plain text. The application uses a dual-layer security approach:

3.1. Master Password & Account Security

Every user establishes a Master Password upon initial login. The Master Password acts as the master cryptographic key for credential data stored within the user's vaults. The Master Password itself is verified locally and session-locked, ensuring that unauthorized individuals cannot view credentials even if they gain physical access to an active browser window.

3.2. Zero-Knowledge Cryptography

Under the Zero-Knowledge principle, Vaultix servers never store or possess user master passwords. When a user saves or updates a password entry, encryption occurs directly inside the user's browser before transmission. The server only stores encrypted data payloads that are unreadable without the user's Master Password.

3.3. Master Password Rotation & Automatic Re-encryption

When a user updates their Master Password, Vaultix automatically decrypts all existing entries owned by the user in memory using the old key, re-encrypts them with the new key, and updates the system vault in a single atomic transaction. This guarantees continuous security without risk of data lock-out.

3.4. Session Inactivity Auto-Lock

To protect against unattended device exposure, Vaultix continuously monitors user interaction. If no user activity (mouse movement, keyboard input, or touch gestures) is detected for 15 minutes, the application automatically locks the vault and terminates the active session.

CRITICAL SECURITY RULE: *Never share your Master Password with anyone. Because Vaultix uses Zero-Knowledge encryption, lost Master Passwords cannot be recovered by system administrators.*

4. User Roles & Access Control Hierarchy

Vaultix enforces strict separation of duties using system roles, vault access levels, and folder permissions.

4.1. System Roles

System Role	Scope	Capabilities & Privileges
Administrator (ADMIN)	Full system oversight	Can create shared vaults, manage team members, send email invitations, update user roles, toggle account active status, and review global login logs.
Standard User (USER)	Vault participant	Can manage their personal vault, create personal credentials, and interact with shared folders/passwords explicitly granted to them.

4.2. Folder Permission Levels

Permission Level	Type	Permitted Actions
Administrator	Full Folder Governance	Full control over folder content, password entries, and ability to manage user folder access permissions.
Full Access	Create & Modify	Add, view, edit, copy, and delete credentials within the designated folder.
Edit Only	Update Content	View existing credentials, copy passwords, and update entry details (cannot delete entries).
Read Only	View & Copy Only	View credential details and copy passwords. All view and copy actions are logged for security compliance.

5. Feature & Module Breakdown

5.1. Personal Vault ('My Vault')

Every Vaultix account includes a dedicated Personal Vault automatically. Personal Vaults are completely private to the individual account owner.

- **Personal Folders:** Organize personal logins into custom subfolders.
- **Color Tags & Labels:** Assign custom color tags and custom text labels to items.
- **Credential Entry Details:** Store site titles, usernames, encrypted passwords, target website URLs, and encrypted text notes.
- **Search & Filter:** Instant client-side searching by name, username, URL, or tag.

5.2. Shared Vaults (Company & Client Vaults)

Shared Vaults enable team collaboration across organization departments or client projects.

- **Company Vaults:** Created by Administrators to group company-wide credentials.
- **Client Vaults:** Dedicated spaces for client accounts, partner credentials, or external projects.
- **Folder-Level Access:** Administrators assign specific team members to shared folders within company vaults with custom access levels.

5.3. Password Generator Engine

Vaultix includes a built-in customizable password generator that produces strong, random passwords to eliminate weak or repeated credentials.

- **Custom Length:** Adjust length anywhere from 8 to 64 characters.
- **Character Options:** Toggle Uppercase (A-Z), Lowercase (a-z), Numbers (0-9), and Special Symbols (!@#\$%^&*).
- **Exclude Ambiguous Characters:** Option to exclude confusing characters such as O and 0, or l, 1, and I.
- **Real-time Entropy Meter:** Displays visual strength indicators based on password entropy.

5.4. Password Security Auditor & Health Score

Vaultix continuously analyzes stored credentials to identify security vulnerabilities across four main criteria:

- **Weak Password Detection:** Identifies short passwords (<8 characters), simple patterns, or missing character diversity.
- **Old Password Detection:** Flags credentials that have remained unchanged for over 180 days.
- **At-Risk Password Detection:** Cross-references credentials against lists of known compromised or breached passwords.

- **Automated Security Score:**Calculates an overall security score from 0 to 100 for the user or organization.

5.5. Executive Security Dashboard

The Security Dashboard delivers high-level visibility into credential health and team usage trends.

- **Summary Cards:**Overall Security Score gauge, Total Passwords count, Company Passwords count, Personal Passwords count, and Deleted Items count.
- **Activity Trend Graphs:**Interactive Recharts line/bar graphs showing added vs. deleted passwords over 7 Days, 30 Days, or 6 Months.
- **Recent Passwords List:**Quick table listing the most recently added or modified credentials.

5.6. Activity Audit Logging

Vaultix records an immutable audit log for compliance and security monitoring. Tracked actions include:

- **Password View Events:**Logged whenever a user reveals a hidden password.
- **Password Copy Events:**Logged whenever a user copies a password to the clipboard.
- **Creation & Edit Events:**Logged when new entries are added or modified.
- **Deletion Events:**Logged when passwords or folders are removed.
- **Sharing Events:**Logged when folders or passwords are shared with team members.

PRIVACY GUARANTEE: *To protect personal privacy, activity log entries associated with a user's Personal Vault are visible only to that user and are hidden from system administrators.*

5.7. Login Activity Monitor

Tracks all account login attempts to protect against credential stuffing or unauthorized access.

- **Audit Parameters:**Records user IP address, browser User-Agent, timestamp, and login status (SUCCESS, FAILED, BLOCKED).
- **Blocked Account Alerts:**Accounts belong to inactive or deactivated users are automatically blocked and flagged.

5.8. Bulk Excel Import & Export Engine

Simplifies onboarding and data backup with structured Excel spreadsheet integration.

- **Bulk Import:**Import up to 500 credentials at once using standardized Excel templates. Imported passwords are automatically encrypted in browser memory prior to storing.
- **Secure Export:**Export vault entries into structured Excel spreadsheets for offline backup or compliance reporting.

5.9. Team Management & User Administration

Administrators can manage the entire organization's user lifecycle from a single control panel.

- **Manual User Creation:**Create user accounts manually with designated roles.
- **Account Deactivation:**Instantly suspend account access by toggling account active status.
- **Email Invitations:**Send formal registration invitations to new employees via automated email with secure 7-day expiration tokens.

5.10. Notifications & Visual Themes

- **In-App Notification Center:**Header notification drawer providing instant alerts for security warnings, password share invites, and login updates.
- **Light / Dark Theme Support:**Full support for Light and Dark UI modes, synced with browser preferences or toggled manually.

6. Step-by-Step User Workflows

Workflow 1: Setting Up Your Vaultix Account

- **Step 1:**Receive an email invitation link or navigate to the registration page.
- **Step 2:**Enter your Full Name, Email Address, and Account Password.
- **Step 3:**On first login, set your Master Password and an optional hint.
- **Step 4:**Your Personal Vault ('My Vault') is automatically generated and ready for use.

Workflow 2: Adding & Managing Credentials

- **Step 1:**Navigate to 'My Vault' or an authorized Shared Vault folder.
- **Step 2:**Click '+ Add Password'.
- **Step 3:**Enter Item Title, Username, Password (or click Generate), URL, Category Folder, and Color Tags.
- **Step 4:**Click Save. The entry is instantly encrypted and saved securely.

Workflow 3: Sharing a Folder with Team Members (Admins)

- **Step 1:**Open the desired Shared Vault folder.
- **Step 2:**Click 'Share Folder' or 'Manage Users'.
- **Step 3:**Enter the team member's email address and select their permission level (Administrator, Full Access, Edit Only, Read Only).
- **Step 4:**Click Confirm. The folder becomes immediately available in the team member's workspace.

Workflow 4: Bulk Importing Credentials from Excel

- **Step 1:**Navigate to the target vault folder and click 'Import Passwords'.
- **Step 2:**Download the sample Excel template or select your existing file containing Name, Username, Password, and URL columns.
- **Step 3:**Upload the spreadsheet. Vaultix will validate rows, encrypt passwords client-side, and import entries.